

UCab — Full-Stack MERN Cab Booking Platform

Solo Developer / Team Member: Shaik Sumiya Zainab | Project ID: N/A (Solo Track)

Phase 7: Project Documentation — Executable Files & Quick-Start Guide

Project Name: Cab Booking (`UCab`)

Project ID: `N/A (Solo Track Submission)`

Team Member / Solo Developer: Shaik Sumiya Zainab

1. System Prerequisites

Before executing the UCab application, ensure that the target evaluation machine has the following software installed:

- Node.js: v18.0.0 or higher (`v20.x` recommended). Verify via terminal: `node -v`
- NPM Package Manager: v9.0.0 or higher. Verify via terminal: `npm -v`
- MongoDB (Optional): Local `mongod` daemon on port `27017` OR cloud Atlas connection string.

(Note: Thanks to our built-in Zero-Config Storage Engine, if MongoDB is not installed, the server will automatically seed and save data locally to `server/db/data.json` without any installation errors or crashes).

2. Directory Structure of Executable Source Code

The runnable source code is provided inside two accessible locations in this repository:

1. `5. Project Development Phase/UCab/`
2. Root directory `UCab/`

3. Step-by-Step Execution Commands

Step 1: Start the Backend REST API Server (Port 8000)

Open a terminal window and execute:

```
# 1. Navigate to the backend server folder
cd UCab/server

# 2. Install required Node modules (Express, Mongoose, JWT, Bcrypt, Multer)
npm install

# 3. Launch the API Server
npm start
```

Expected Console Output:

```
> ucab-backend@1.0.0 start
> node server.js

-----
& p MongoDB local engine not found on port 27017 (or connected successfully)
'( Automatic Zero-Config MERN Fallback Activated:
'( All Mongoose Schemas & API endpoints will transparently persist data to server/db/
data.json!
-----
Server running on port 8000 Ø=þ€
Seeded/loaded 6 cars from automatic JSON persistence store.
```

Step 2: Start the React + Vite Frontend Web App (Port 5173 / 5174)

Open a second terminal window and execute:

```
# 1. Navigate to the frontend client folder
cd UCab/client
```

```
# 2. Install React, Vite, Router, and Icon packages
npm install
# 3. Launch the Vite Development Server
npm run dev
```

Expected Console Output:

```
VITE v5.4.1  ready in 632 ms
'œ Local:   http://localhost:5173/   (or http://localhost:5174/)
'œ Network: use --host to expose
```

4. Pre-Loaded Evaluation Demo Accounts

To test all user and admin workflows immediately without going through manual registration forms, use our built-in 1-Click Demo accounts:

- Demo Rider Account:
- Email: `sumiya@ucab.com` | Password: `password123`
- *(Action: Click the button labelled "&j One-Click Demo Rider Login" on `/login`)*
- Demo Super Admin Account:
- Email: `admin@ucab.com` | Password: `adminpassword`
- *(Action: Click the button labelled "Ø=Ðáp One-Click Super Admin Login" on `/admin/login`)*